

Course Report

Introductory Bioinformatics

Assignment 1

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Part I

Assignment 1

1 Report Overview

This report presents the work completed for Assignment 1 of the Introductory Bioinformatics course in Fall 2025. The assignment contains three internal task groups. Part A covers downloading and inspecting the sequencing reads and evaluating their quality. Part B covers de novo genome assembly with SPAdes and assembly evaluation with QUAST. Part C covers mapping the reads to three different reference genomes and comparing the resulting alignment statistics. These task groups belong only to Assignment 1 and should not be confused with the separate Assignment 2 or the independent final project.

The original project files were uploaded to the Google Drive folder provided in the submitted report:

https://drive.google.com/drive/folders/1jU0gyFWAo59Wp60pbi0eYJvXJmrQZ_gR

2 Part A: Read Retrieval and Quality Assessment

2.1 Downloading the sequencing data

We first downloaded the SRA Toolkit for Windows from GitHub. We then used `fasterq-dump` to download accession SRR36584168 from the Sequence Read Archive [2]. The command was:

```
& ".\fasterq-dump.exe" SRR36584168 --split-files --threads 6 --outdir
"D:\python\bioinformatics\hw2\data"
```

The `-split-files` option was included so that paired-end data would be separated if the accession contained paired reads. The result showed that the dataset was single-end. One FASTQ file of approximately 1.49 GB was placed in the data directory.

2.2 Programmatic inspection with Biopython

We used Biopython in the script `scripts/parta_sec2.py` to answer the six questions in this section. The main results are summarized in Table 1.

Table 1: Results obtained from the Biopython inspection script.

Item	Result
I	Number of reads: 2,421,219
II	First-read identifier: SRR36584168.1
III	Total occurrences of motif GCGATCAACTCGCTGG, with overlapping occurrences counted: 7,499
IV	The first 1,000 reads were written to <code>data/first_1000.fastq.gz</code> .
V	The quality boxplot was saved as <code>plots/per_base_quality_boxplot.png</code> .
VI	The read-length density plot was saved as <code>plots/read_length_density.png</code> .

The sequence of the first read was:

```
CGTTCATGACCGAAGCGGCGATCAACTCGCTGGCCGGCGAAGCGCTGAACGACATG
TATGCGCAGTGCCAAACCAGCACTGGTCTGAATCCGGCATTCCAGGGGAATGGCGC
TCAGAGTGAAATCCTTTCTATGGAGGCTCCTGAATAATGGCTCTCGCACCTCGTTTCC
ATACCGTAATCGCGGGCCCGCCGCAAGAATGACCCGCAGGTCATTGAAGCAATC
ATGGCGGCGGCAGTGAAGCCTGGGT
```

The Phred quality values reported for this read were:

```
[37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37,
37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37,
37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37,
37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37,
37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37,
37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37,
37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37,
37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37,
37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37,
37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37,
37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37,
37, 37, 25, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37,
37, 37, 37, 37, 37, 37, 25, 37, 37, 11, 37, 37, 37, 37, 37, 37,
37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 25, 37, 37, 37,
37, 37, 37, 37, 37, 25, 37, 37, 37, 37, 37, 37, 37, 37, 37,
37, 37, 37, 37, 37, 37, 37, 37]
```

Figures 1 and 2 show the plots saved by the script.

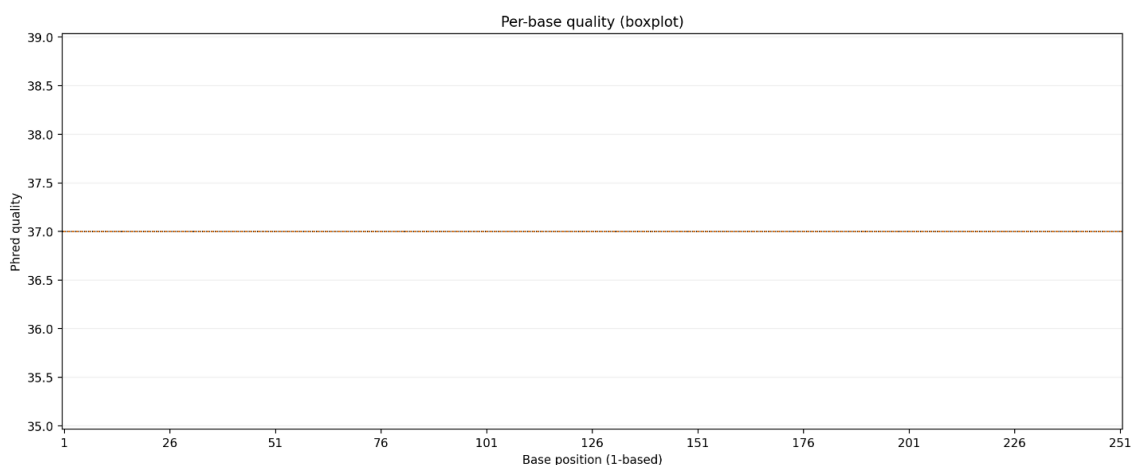


Figure 1: Per-base Phred quality boxplot for the sequencing reads.

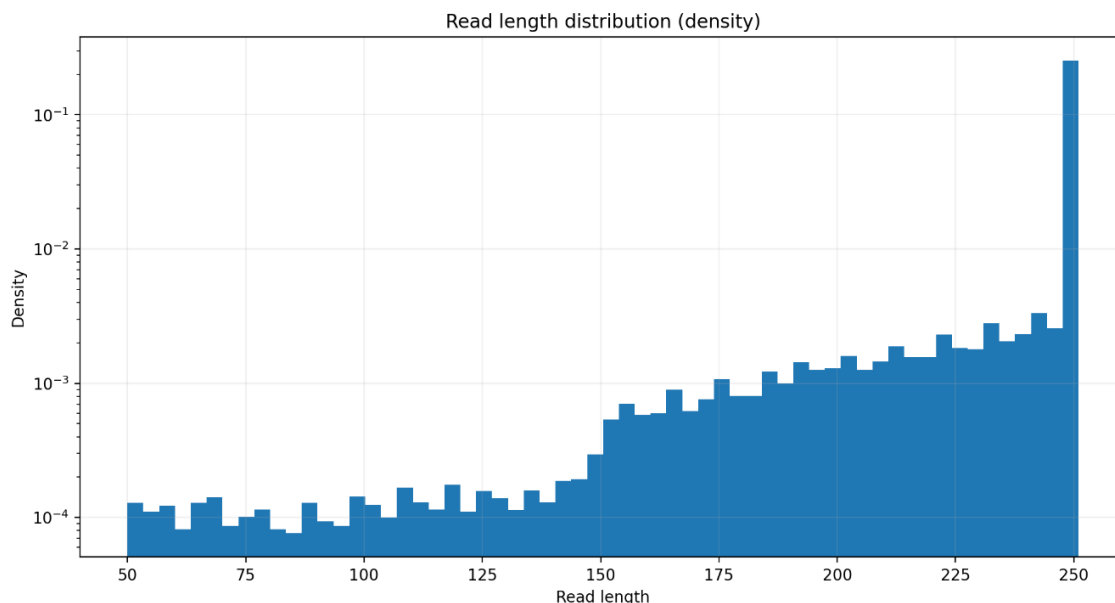


Figure 2: Distribution of read lengths in the downloaded FASTQ file.

2.3 FastQC analysis

After downloading FastQC, we ran it on `SRR36584168.fastq`. The generated HTML and ZIP output files were stored in the `qc` directory. The command was:

```
.\run_fastqc.bat -o "D:\python\bioinformatics\hw2\qc" -t 6
"D:\python\bioinformatics\hw2\data\SRR36584168.fastq"
```

This command specifies the executable, the output location, six threads, and the input FASTQ file. The basic statistics are reproduced as an actual LaTeX table in Table 2.

Table 2: FastQC basic statistics for `SRR36584168.fastq`.

Measure	Value
Filename	<code>SRR36584168.fastq</code>
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total sequences	2,421,219
Total bases	591.5 Mbp
Sequences flagged as poor quality	0
Sequence length	50–251 bp
GC content	53%

The module-level results are shown in Table 3.

Table 3: FastQC module summary.

Module	Status
Basic statistics	Pass
Per-base sequence quality	Pass
Per-tile sequence quality	Pass
Per-sequence quality scores	Pass
Per-base sequence content	Pass
Per-sequence GC content	Warning
Per-base N content	Pass
Sequence length distribution	Warning
Sequence duplication levels	Fail
Overrepresented sequences	Pass
Adapter content	Pass

Overall, the data quality was suitable. Per-base sequence quality, per-tile sequence quality, and per-sequence quality scores all passed. This indicates high base quality across the reads and no clear systematic error pattern across sequencing tiles. Adapter content and overrepresented sequences also passed, so there was no strong sign of adapter contamination or unusually frequent sequences.

Two warnings and one failure were reported. Sequence length distribution received a warning because the read lengths varied from 50 to 251 bp. Per-sequence GC content also received a warning, which may be related to base-composition bias or characteristics of the sequenced genome. The sequence duplication level failed, showing high duplication. Possible explanations include very high coverage, a small phage genome, or library preparation effects such as PCR duplication. However, because the base-quality modules passed and no adapter contamination was detected, the reads were considered usable for the following steps.

3 Part B: De Novo Genome Assembly and Evaluation

In this section, we performed de novo genome assembly on the short-read data with SPAdes. We then evaluated the assembly with QUAST and compared it with the initially supplied reference genome, MG770379. The output files for this section are stored in the `assembly` directory. All commands in this part were executed inside Windows Subsystem for Linux (WSL).

3.1 Preparing the software environment

3.1.1 Installing Miniforge

We did not initially have Conda inside WSL. Although installing Miniforge was not itself part of the assignment, it was necessary for creating the environment used to install SPAdes and QUAST. We used the following commands:

```
cd ~
wget -O Miniforge3.sh https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-Linux-x86_64.sh
bash Miniforge3.sh -b -p $HOME/miniforge3
echo 'source $HOME/miniforge3/etc/profile.d/conda.sh' >> ~/.bashrc
source ~/.bashrc
conda --version
```

The commands perform the following operations:

1. Move the terminal to the home directory.
2. Download the Miniforge installer script from GitHub with `wget`.
3. Run the installer non-interactively with `-b` and install it under the home directory.
4. Add the Conda initialization command to `~/.bashrc`, so Conda becomes available automatically in new terminal sessions.
5. Reload `~/.bashrc` in the current terminal.
6. Check the Conda version to confirm that installation succeeded.

3.1.2 Installing SPAdes and QUAST

We created a Conda environment called `asm` and installed SPAdes and QUAST from the Conda-Forge and Bioconda channels:

```
conda create -n asm -c conda-forge -c bioconda spades quast -y
conda activate asm
spades.py --version
quast.py --version
```

The first command creates the environment and installs both tools. The second activates the environment, and the final two commands check that the tools were installed correctly.

3.2 Running SPAdes

We ran SPAdes using the single-end FASTQ file:

```
spades.py \
-s /mnt/d/python/bioinformatics/hw2/data/SRR36584168.fastq \
-o /mnt/d/python/bioinformatics/hw2/assembly/spades_out \
--careful -t 6 -m 16
```

The `-s` option specifies the single-end reads, and `-o` specifies the output directory. The `-careful` mode was used to reduce errors. Six threads were used for parallel processing, and the maximum memory allocation was set to 16 GB.

The most important output files in `spades_out` were:

- `contigs.fasta`: the final assembled contigs;
- `scaffolds.fasta`: the assembled scaffolds;
- `spades.log`: the execution log and messages produced during assembly.

At this stage, SPAdes performed de novo assembly on the WGS reads and produced genomic contigs from the single-end reads in `SRR36584168.fastq`.

3.3 QUAST evaluation with the initially supplied reference

We downloaded the reference genome given in the assignment and stored it as `data/reference.fasta`. We then ran QUAST as follows:

```
quast.py \
/mnt/d/python/bioinformatics/hw2/assembly/spades_out/contigs.fasta \
-r /mnt/d/python/bioinformatics/hw2/data/reference.fasta \
-o /mnt/d/python/bioinformatics/hw2/assembly/quast_out \
--threads 6
```

The QUAST results were stored in `quast_out`, and the main results could be viewed in `report.html`. The reference-independent statistics are reproduced in Table 4.

Table 4: QUAST statistics for the SPAdes contigs without reference-dependent measures.

Measure	Value
Number of contigs	22
Number of contigs (≥ 0 bp)	54
Number of contigs ($\geq 1,000$ bp)	19
Number of contigs ($\geq 5,000$ bp)	9
Number of contigs ($\geq 10,000$ bp)	3
Number of contigs ($\geq 25,000$ bp)	1
Number of contigs ($\geq 50,000$ bp)	0
Largest contig	30,788 bp
Total length	139,526 bp
Total length (≥ 0 bp)	149,588 bp
Total length ($\geq 1,000$ bp)	137,675 bp
Total length ($\geq 5,000$ bp)	108,725 bp
Total length ($\geq 10,000$ bp)	69,143 bp
Total length ($\geq 25,000$ bp)	30,788 bp
Total length ($\geq 50,000$ bp)	0 bp
N50	8,023 bp
N90	3,002 bp
auN	14,749
L50	4
L90	14
GC content	50.98%

All reported assembly evaluations were based on the 22 contigs longer than 500 bp. The total assembly length was 139,526 bp, and the largest contig was 30,788 bp. The continuity measures were $N50 = 8,023$ bp and $L50 = 4$, while $N90 = 3,002$ bp and $L90 = 14$. Therefore, four large contigs were sufficient to cover 50% of the assembly length, but fourteen contigs were needed to cover 90%. This suggests moderate continuity, but the assembly was still not fully continuous.

The assembly GC content was 50.98%, while the initially supplied reference had a length of 156,801 bp and a GC content of 46.14%. QUAST reached the contig-alignment stage but found no valid alignment between the assembled contigs and this reference. As a result, it could not calculate reference-dependent measures such as genome fraction, NG statistics, mismatch and indel rates, or reference-based misassemblies. Only the reference-independent statistics in Table 4 were reliable for this comparison.

The most likely explanation was that the supplied reference did not match the actual content of the reads and assembly. For example, it could belong to a different isolate or phage, or the reads could contain a large proportion of non-target sequences. The difference in GC content, 50.98% for the assembly compared with 46.14% for the reference, also supported this interpretation. Another possibility was that the sequenced sample was mixed with host DNA or contamination. In that situation, SPAdes could assemble the dominant component, while the resulting contigs would not necessarily represent the expected reference.

3.4 Additional check with a BLAST-selected reference

To investigate the mismatch further, we searched the first contig with NCBI BLASTN. We selected one of the matching genomes and downloaded accession CP052431.1 [1]. It was stored in the `data` directory as `blast_found_reference.fasta`. We then repeated QUAST with this reference. The results were stored under `assembly/quast_out_for_blast_found_reference`. This additional check was not required by the assignment, but we performed it to see how the reference-dependent statistics changed when a reference with some similarity to the contigs was used.

Table 5: QUAST alignment-based statistics with the BLAST-selected reference, CP052431.1.

Measure	Value
Genome fraction	1.668%
Duplication ratio	1.018
Largest alignment	26,021 bp
Total aligned length	93,896 bp
NG50 / NG90	Not reported
auNG	372
NA50	4,767 bp
NA90	Not reported
auNA	8,254.2
NGA50 / NGA90	Not reported
auNGA	208.2
LA50	7
Other LG and LGA measures	Not reported

Alignment succeeded with this reference, so QUAST could produce some reference-dependent values. However, the genome fraction was only 1.668%. Therefore, this result showed partial similarity, not a complete reference match.

3.5 Evaluation with the later supplied `ref_genome.fasta`

A new reference called `ref_genome.fasta` was later provided and placed in the `data` directory. We ran QUAST again with the same contigs and stored the output in `assembly/quast_out_for_ref_genome`. Table 6 reproduces the reported alignment statistics.

Table 6: QUAST alignment-based statistics with `ref_genome.fasta`.

Measure	Value
Genome fraction	94.873%
Duplication ratio	1.023
Largest alignment	26,189 bp
Total aligned length	138,259 bp
NG50 / NG90	8,023 bp / 3,002 bp
auNG	14,440
NA50 / NA90	8,023 bp / 2,351 bp
auNA	12,229
NGA50 / NGA90	7,114 bp / 1,916 bp
auNGA	11,973
LG50 / LG90	4 / 14
LA50 / LA90	4 / 17
LGA50 / LGA90	5 / 18

The assembly length for contigs of at least 500 bp was 139,526 bp, compared with a reference length of 142,511 bp. The contigs covered approximately 94.87% of the reference. The assembly GC content was 50.98%, which was close to the reference GC content of 51.11%. Only one contig was completely unaligned, and its length was 651 bp. These results suggest that the amount of contamination or sequence outside this reference was limited.

The assembly was still fragmented, with 22 contigs of at least 500 bp, an N50 of 8,023 bp, and a largest contig of 30,788 bp. QUAST reported five structural misassemblies classified as translocations, showing that some connections were incorrect. At the nucleotide level, the report described approximately 111 mismatches and 8.68 indels per 100 kbp. Overall, the SPAdes result had suitable reference coverage and nucleotide composition, although it was fragmented and contained several structural errors.

3.6 Part B conclusion

The first supplied reference did not align to the assembly and was not suitable for reference-based evaluation. The BLAST-selected genome allowed a partial comparison, but its genome fraction was only 1.668%. The later supplied `ref_genome.fasta` matched the assembly much more closely, with 94.873% genome fraction and similar GC content. This sequence of checks showed why the choice of reference strongly affected the QUAST interpretation.

4 Part C: Mapping Illumina Reads to Reference Genomes

Because the initially supplied reference did not match the assembled genome, we first used BLAST on an assembled contig to identify a more suitable reference. This search produced `blast_found_reference.fasta`, corresponding to accession CP052431.1. We used it for the first mapping analysis. We then repeated the same analysis with the original supplied reference and with the later supplied `ref_genome.fasta`. This allowed us to compare the behavior of the reads against three different references.

4.1 Mapping to the BLAST-selected reference

4.1.1 BWA indexing and alignment

The reads were single-end Illumina reads. We first indexed the reference with BWA and then mapped the reads with `bwa mem`:

```
bwa index blast_found_reference.fasta
bwa mem blast_found_reference.fasta SRR36584168.fastq > alignment.sam
```

The second command produced `alignment.sam`, which contains the read alignments.

4.1.2 Inspecting the SAM file

We inspected the first lines with:

```
cd ~/PART_C
head alignment.sam
```

The SAM file begins with header lines marked by `@`. These lines describe the reference sequences and the mapping program. The remaining alignment lines contain the read identifier, mapping flag, reference name, mapping position, mapping quality, CIGAR string, read sequence, and base-quality values.

The main fields are:

- **Read identifier:** the name assigned to the read.
- **Flag:** the alignment status and properties of the read.
- **Reference name:** the chromosome, contig, or genome to which the read was mapped.
- **Position:** the location where the read alignment begins on the reference.
- **Mapping quality:** a confidence score for the alignment, where larger values indicate higher confidence.
- **Other fields:** the CIGAR string, read sequence, and base-quality scores.

SAM also contains fields such as mate reference, mate position, and insert size for paired-end data. These fields were not needed here because this dataset was single-end.

Example header and alignment lines were:

```
@SQ SN:CP052431.1 LN:5532256
@PG ID:bwa PN:bwa VN:0.7.17-r1188 CL:bwa mem blast_found_reference.fasta SRR36584168.
fastq
SRR36584168.4 0 CP052431.1 2356521 60 4S34M164S * 0 0 CGAACGATGACAACTTT... FFFFFFFFFF
```

In the `@SQ` line, `SN` is the reference sequence name and `LN` is its length. The `@PG` line records the program, BWA-MEM, its version, and the command used. The alignment line shows that read `SRR36584168.4` mapped to `CP052431.1` at position 2,356,521 with mapping quality 60. The CIGAR string `4S34M164S` means that 4 bases at the beginning were soft-clipped, 34 bases were aligned, and 164 bases at the end were soft-clipped.

QNAME	FLAG	RNAME	POS	MAPQ	CIGAR	SEQ
SRR36584168.1	4	* 0	0	*	* 0	CGTTTCATGACCGAAGCGGCGATCACTCGCTGGCCGGCGAAGCGCTGAACGACATGTATGCGCAGTGCCAAACC/
SRR36584168.2	4	* 0	0	*	* 0	CCGCCACCAACAATAATGCGCTGACTGCGGCCATTCCTCAACAAACACAGAGGAAGTATCAAAATGGAGA/
SRR36584168.3	4	* 0	0	*	* 0	TCAATAAGGCGGTGCGCGAGACAAAGCTGACGGCAAGCGCTGATTAATGCTGCTGATGAGATGATCGCCGGC/
SRR36584168.4	0	CP052431.1	2356521	60	4S34M164S	* 0
SRR36584168.5	4	* 0	0	*	* 0	0 BCGAACGATGACAACTTTGCTTACTTCGCTGCGCTGGCGGTGCCGATCCAGTGTCTATTGCTATTAA/
SRR36584168.6	4	* 0	0	*	* 0	ATCCTTTTATGCGGACTCAGCATCTTTAAGTGACAAATACTCCCTATTCTTTGTTTTCCCTTTATGGATA/
SRR36584168.7	4	* 0	0	*	* 0	GACAACATCACCACCCGAGTCTTCGCAACGACGAAGCAACGGCTGCTTGTATGACCTGCTCGGTCTCGGTAC/
SRR36584168.8	4	* 0	0	*	* 0	CGAAAAACCATGCGGGCGCTCTACTCTGCTCTGTTTGTGATTGGGCTGGAGTCCGCGGTGCTGTTCTCATCGA/
SRR36584168.9	4	* 0	0	*	* 0	CACGAAACGTCACTGGCGCCGGGAACATCATCTCTCTGAGAGCTTTATCGGAGACGACACGCTGCTGACAG/
SRR36584168.10	4	* 0	0	*	* 0	

Figure 3: Preview of the SAM output after mapping reads to the BLAST-selected reference CP052431.1.

4.1.3 Converting, sorting, and indexing the alignment

We converted the SAM file to BAM, sorted it, and created an index:

```
samtools view -bS alignment.sam > alignment.bam
samtools sort alignment.bam -o alignment_sorted.bam
samtools index alignment_sorted.bam
```

The sorted BAM file was used for IGV visualization and for the alignment statistics in the following sections.

4.1.4 IGV visualization

We loaded the reference genome and `alignment_sorted.bam` into IGV. The original report selected the region `CP052431.1:2,356,504-2,356,578` to examine the read coverage and alignment quality.

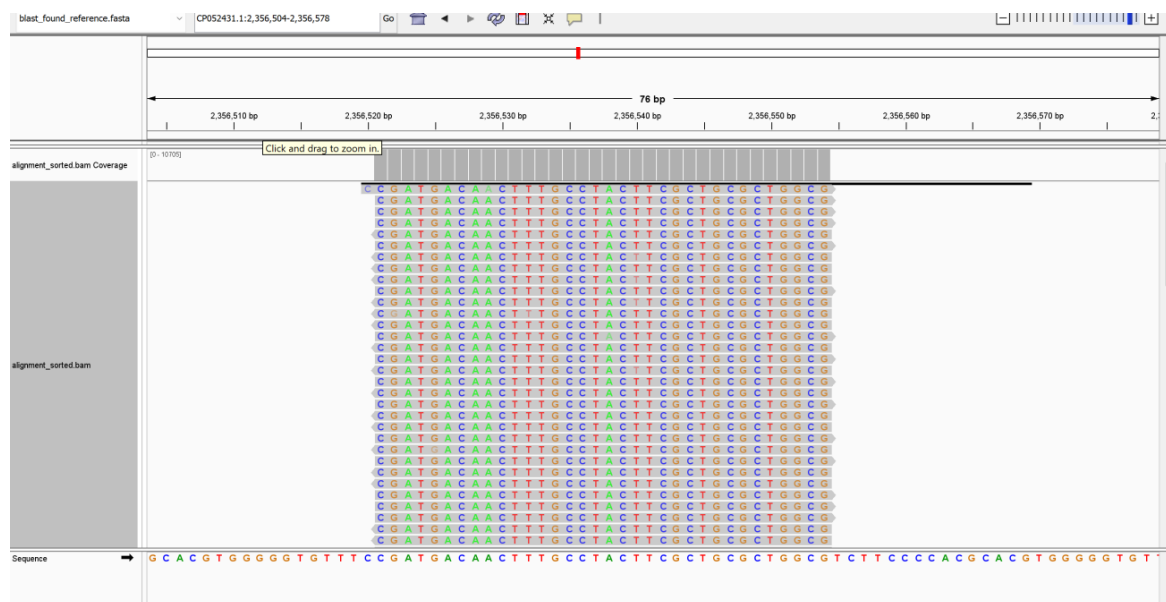


Figure 4: IGV visualization of reads aligned to CP052431.1 in the selected region.

The selected region contains aligned reads with local coverage. Some reads show small mismatches and soft-clipped regions. This behavior can occur in Illumina short-read alignment. The visualization confirmed that mapping to this reference had been performed correctly, although it did not show that the complete dataset matched the reference.

4.1.5 Mapped-read percentage

We calculated the mapping statistics with:

```
samtools flagstat alignment_sorted.bam
```

```
2421504 + 0 in total (QC-passed reads + QC-failed reads)
2421219 + 0 primary
0 + 0 secondary
285 + 0 supplementary
0 + 0 duplicates
0 + 0 primary duplicates
178520 + 0 mapped (7.37% : N/A)
178235 + 0 primary mapped (7.36% : N/A)
0 + 0 paired in sequencing
0 + 0 read1
0 + 0 read2
0 + 0 properly paired (N/A : N/A)
0 + 0 with itself and mate mapped
0 + 0 singletons (N/A : N/A)
0 + 0 with mate mapped to a different chr
0 + 0 with mate mapped to a different chr (mapQ>=5)
```

Figure 5: Samtools flagstat output for reads mapped to CP052431.1.

The output reported 2,421,504 total records, consisting of 2,421,219 primary alignments and 285 supplementary alignments. A total of 178,520 records were mapped, giving a reported mapping percentage of 7.37%. The primary mapped count was 178,235, or 7.36%. This distinction explains why the total shown by flagstat differs slightly from the original FASTQ read count and why two closely related mapped counts appear in the report. The low percentage shows that CP052431.1 represents only a subset of the read content, even though it was one of the references found from the initial BLAST search.

4.1.6 Read depth

Read depth is the number of reads covering each reference position. We generated a depth file and calculated the mean depth with:

```
samtools depth alignment_sorted.bam > depth.txt
awk '{sum+=$3} END {print "Mean read depth:", sum/NR}' depth.txt
```

Each line of depth.txt contains the reference name, position, and depth. An example from the report was:

```
CP052431.1 19885 1
CP052431.1 19886 1
CP052431.1 19887 1
CP052431.1 19888 1
CP052431.1 19889 1
CP052431.1 19890 1
CP052431.1 19891 1
CP052431.1 19892 1
CP052431.1 19893 1
CP052431.1 19894 1
```

The mean read depth was 91.5511, which was reported in the discussion as approximately 91.55.

Mean read depth: 91.5511

Figure 6: Mean read depth calculated for reads mapped to CP052431.1.

4.2 Interpreting the CIGAR format

CIGAR stands for Compact Idiosyncratic Gapped Alignment Report. It describes how a sequenced read aligns to a reference, including aligned regions, mismatches, insertions, deletions, clipping, and skipped regions. Table 7 lists the symbols used in the original report.

Table 7: Meaning of common CIGAR symbols.

Symbol	Meaning
M	Alignment match, which may be a base match or mismatch
=	Exact match
X	Mismatch
I	Insertion in the read
D	Deletion from the reference
S	Soft clipping, where bases are kept but not aligned
N	Skipped region in the reference

The three examples were interpreted as follows:

1. 29S21=1X25=: 29 soft-clipped bases, followed by 21 exact matches, one mismatch, and 25 exact matches.
2. 20M2I1M1D10M: 20 aligned bases, a two-base insertion in the read, one aligned base, a one-base deletion from the reference, and 10 aligned bases.
3. 5M10N25M: 5 aligned bases, a 10-base skipped region in the reference, and 25 aligned bases.

All operations occur in the order shown in the CIGAR string.

4.3 Repeating the analysis with the original supplied reference

We repeated the complete workflow with `reference.fasta`:

```
bwa index reference.fasta
bwa mem reference.fasta SRR36584168.fastq > alignment_reference.sam
samtools view -S -b alignment_reference.sam > alignment_reference.bam
samtools sort alignment_reference.bam -o alignment_reference_sorted.bam
samtools index alignment_reference_sorted.bam
```

The SAM preview is shown in Figure 7.

```
@SQ      SN:MG770379.1  LN:156801
@PG      ID:bwa      PN:bwa      VN:0.7.17-r1188  CL:bwa mem reference.fasta SRR36584168.fastq
SRR36584168.1  4      *      0      0      *      *      0      0      CGTTCATGACCGAAGCGGCGATCAACTCGCTGGCC
SRR36584168.2  4      *      0      0      *      *      0      0      CCGCCACCAACAATAATGCCGCTGACTGGCGGC
SRR36584168.3  4      *      0      0      *      *      0      0      TCAATAAGGCGGTGCGCGAGACAAAGCCTGACGGC
SRR36584168.4  4      *      0      0      *      *      0      0      CGAACGATGACAACTTTGCCCTACTTCGCTGCGCTG
SRR36584168.5  4      *      0      0      *      *      0      0      ATCCTTTATTGCCGACTCAGCATCTTTAAGTGACA
SRR36584168.6  4      *      0      0      *      *      0      0      GACAACATCACCACCGAGTCTTCGCAACGACGAC
SRR36584168.7  4      *      0      0      *      *      0      0      CGAAAAACCATGCGGGCGCTCTACTCTGCTGTT
SRR36584168.8  4      *      0      0      *      *      0      0      CACGAAACGTCCTGGCGCCGGGAACATCATCTCC
```

Figure 7: SAM preview after attempting to map the reads to the original supplied reference.

We could not find a 200-base-pair region in IGV containing aligned reads or visible coverage.

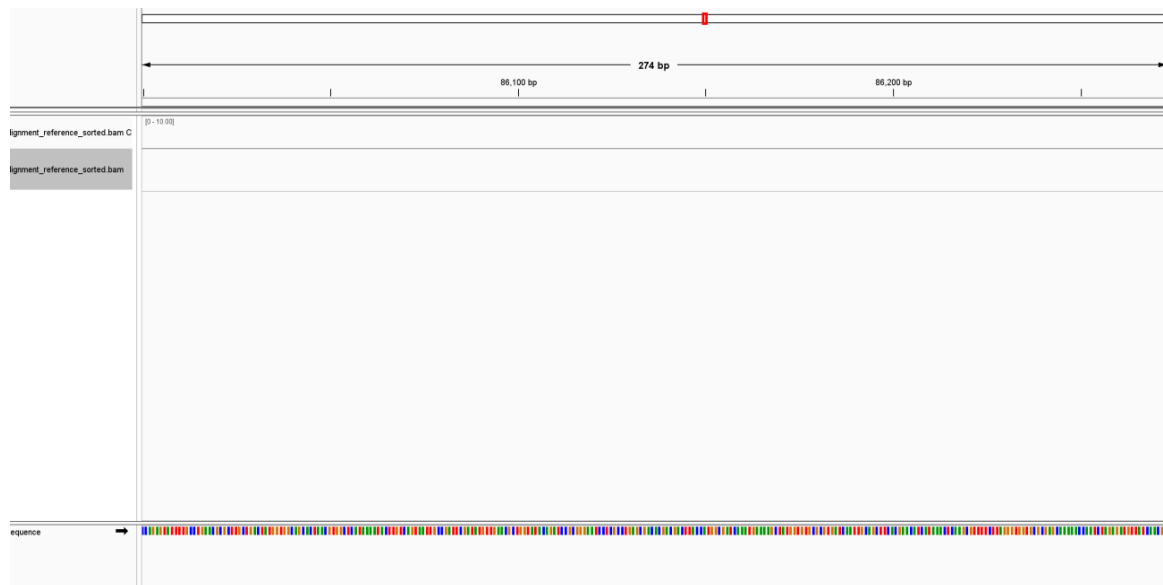


Figure 8: IGV view showing no read alignments to the original supplied reference.

The flagstat output confirmed that none of the reads mapped to this reference.

```
2421219 + 0 in total (QC-passed reads + QC-failed reads)
2421219 + 0 primary
0 + 0 secondary
0 + 0 supplementary
0 + 0 duplicates
0 + 0 primary duplicates
0 + 0 mapped (0.00% : N/A)
0 + 0 primary mapped (0.00% : N/A)
0 + 0 paired in sequencing
0 + 0 read1
0 + 0 read2
0 + 0 properly paired (N/A : N/A)
0 + 0 with itself and mate mapped
0 + 0 singletons (N/A : N/A)
0 + 0 with mate mapped to a different chr
0 + 0 with mate mapped to a different chr (mapQ>=5)
```

Figure 9: Samtools flagstat output showing a 0.00% mapping rate for the original supplied reference.

This 0% result agrees with the previous QUAST analysis and shows that the reads and assembled genome differ greatly from the original reference supplied in the assignment. Because no reads were aligned, the depth was zero across the reference and `depth_reference.txt` was empty. Attempting to calculate the mean with the same AWK command produced:

```
awk: cmd. line:1: fatal: division by zero attempted
```

The division-by-zero error occurred because the depth file contained no rows.

4.4 Repeating the analysis with ref_genome.fasta

Finally, we repeated the same BWA and Samtools steps using `ref_genome.fasta`. The SAM header contained one `@SQ` line for each contig in this reference. For example:

```
@SQ SN:k141_29 LN:47524
@PG ID:bwa PN:bwa VN:0.7.17-r1188 CL:bwa mem ref_genome.fasta SRR36584168.fastq
SRR36584168.1 16 k141_29 20702 60 251M
```

The first line states that contig `k141_29` has a length of 47,524 bp. The `@PG` line records BWA version 0.7.17 and the command used. In the alignment line, read `SRR36584168.1` mapped to the reverse strand of `k141_29` at position 20,702 with mapping quality 60. The CIGAR string `251M` means that all 251 read bases were aligned. The tags `NM:i:0` and `MD:Z:251` in the full record indicated zero mismatches and a complete match for that read.

```
alignment_ref.sam X
C: > Users > REC > Desktop > PART_C > alignment_ref.sam
1 @SQ SN:k141_8 LN:307
2 @SQ SN:k141_12 LN:368
3 @SQ SN:k141_30 LN:319
4 @SQ SN:k141_14 LN:374
5 @SQ SN:k141_0 LN:315
6 @SQ SN:k141_4 LN:328
7 @SQ SN:k141_18 LN:388
8 @SQ SN:k141_26 LN:614
9 @SQ SN:k141_10 LN:350
10 @SQ SN:k141_6 LN:406
11 @SQ SN:k141_1 LN:370
12 @SQ SN:k141_9 LN:579
13 @SQ SN:k141_2 LN:496
14 @SQ SN:k141_22 LN:355
15 @SQ SN:k141_5 LN:465
16 @SQ SN:k141_7 LN:305
17 @SQ SN:k141_11 LN:386
18 @SQ SN:k141_3 LN:327
19 @SQ SN:k141_13 LN:1987
20 @SQ SN:k141_19 LN:1879
21 @SQ SN:k141_20 LN:3479
22 @SQ SN:k141_25 LN:4605
23 @SQ SN:k141_15 LN:4318
24 @SQ SN:k141_16 LN:5745
25 @SQ SN:k141_21 LN:2544
26 @SQ SN:k141_27 LN:6292
27 @SQ SN:k141_24 LN:7128
28 @SQ SN:k141_17 LN:2600
29 @SQ SN:k141_23 LN:6981
30 @SQ SN:k141_28 LN:14175
31 @SQ SN:k141_31 LN:26202
32 @SQ SN:k141_29 LN:47524
33 @PG ID:bwa PN:bwa VN:0.7.17-r1188 CL:bwa mem ref_genome.fasta SRR36584168.fastq
34 SRR36584168.1 16 k141_29 20702 60 251M * 0 0 ACCCAGGCTTCACTGCCGCCCATGATTGCTTCAATGACCTGCGGGTCATTCTTGCGG
35 SRR36584168.2 16 k141_29 38081 60 251M * 0 0 ACTCAAGAACAGCCAGTAACATCGCCATTTTCGGCACCCACGATGATTTCCAGCGCGTA
```

Figure 10: SAM preview for reads mapped to `ref_genome.fasta`.

We loaded `ref_genome.fasta` and the sorted BAM file, `alignment_ref_sorted.bam`, into IGV. Figure 11 shows continuous read coverage along contig `k141_29`.



Figure 11: IGV visualization showing dense and continuous read coverage on contig k141_29.

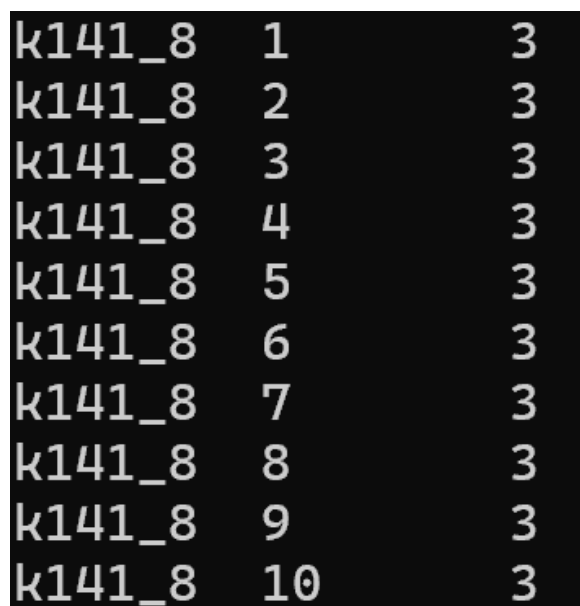
The flagstat output reported a 99.94% mapping rate.

```
2429116 + 0 in total (QC-passed reads + QC-failed reads)
2421219 + 0 primary
0 + 0 secondary
7897 + 0 supplementary
0 + 0 duplicates
0 + 0 primary duplicates
2427687 + 0 mapped (99.94% : N/A)
2419790 + 0 primary mapped (99.94% : N/A)
0 + 0 paired in sequencing
0 + 0 read1
0 + 0 read2
0 + 0 properly paired (N/A : N/A)
0 + 0 with itself and mate mapped
0 + 0 singletons (N/A : N/A)
0 + 0 with mate mapped to a different chr
0 + 0 with mate mapped to a different chr (mapQ>=5)
```

Figure 12: Samtools flagstat output for ref_genome.fasta, showing 99.94% mapped records.

This run contained 2,429,116 total alignment records, including 2,421,219 primary records and 7,897 supplementary records. A total of 2,427,687 records were mapped, and 2,419,790 primary records were mapped. Both reported mapping percentages were 99.94%.

The depth file again contained the contig name, position, and read depth. Example rows are shown in Figure 13.



k141_8	1	3
k141_8	2	3
k141_8	3	3
k141_8	4	3
k141_8	5	3
k141_8	6	3
k141_8	7	3
k141_8	8	3
k141_8	9	3
k141_8	10	3

Figure 13: Example depth rows for contig `k141_8` in `ref_genome.fasta`.

The mean read depth for this reference was 4,147.63. The high mapping percentage, continuous IGV coverage, and large mean depth all support a strong match between the reads and `ref_genome.fasta`.

4.5 Comparison of the three references

Table 8 summarizes the mapping results.

Table 8: Comparison of read mapping against the three reference genomes.

Reference		Mapping rate		Mean depth		Interpretation
Original <code>reference.fasta</code>	supplied	0.00%		0;	depth file	The reference does not match the sequenced sample.
BLAST-selected	CP052431.1	7.37% overall; 7.36% primary		91.5511		Some local similarity exists, but the reference represents only a subset of the reads.
Later <code>ref_genome.fasta</code>	supplied	99.94%		4,147.63		The reads match this reference very strongly.

The original reference produced no mappings. The BLAST-selected reference allowed a limited analysis, but only 7.37% of records mapped. In contrast, `ref_genome.fasta` produced a 99.94% mapping rate. Many alignments had 251M, NM:i:0, and MAPQ=60, while IGV showed continuous and dense coverage. These results confirm that the final reference is much more consistent with the sequencing data.

4.6 Assignment 1 conclusion

The sequencing reads had generally good base quality and no strong adapter contamination, although their length distribution varied and the duplication level was high. SPAdes produced a

139,526 bp assembly with 22 contigs of at least 500 bp and an N50 of 8,023 bp. The first reference did not match the assembly or the reads. A BLAST-selected reference produced only partial agreement. The later supplied `ref_genome.fasta` matched the data much better, with 94.873% QUAST genome fraction and a 99.94% read-mapping rate. Therefore, the final interpretation depends strongly on using the correct reference genome.

References

- [1] National Center for Biotechnology Information. Nucleotide record CP052431.1. NCBI Nucleotide Database. Accession and URL stated in the original report; organism metadata should be verified before final submission.
- [2] National Center for Biotechnology Information. Sequence Read Archive record SRR36584168. NCBI Sequence Read Archive. Accession number stated in the original report; bibliographic metadata should be verified before final submission.